

Bits 11:2 Reserved, must be kept at reset value.

Bit 1 **FORMAT**: Base address register format
1: Arm[®] debug interface v5.

Bit 0 **ENTRYPRESENT**: Debug component present status bit
This bit indicates that debug components are present on the access port bus.

0: Debug components are not present (AP1)
1: Debug components are present (AP0, AP2)

Access port identification register (AP_IDR)

Address offset: 0xFC

Reset value: 0x8477 0001 (AP0 and AP1), 0x5477 0002 (AP2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REVISION[3:0]				JEDEC BANK[3:0]				JEDEC CODE[6:0]						MEMAP	
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IDENTITY[7:0]							
								r	r	r	r	r	r	r	r

Bits 31:28 **REVISION[3:0]**: Arm core revision
0x5: r1p0 (AP2)
0x8: r0p9 (AP0 and AP1)

Bits 27:24 **JEDEC BANK[3:0]**: JEDEC bank
0x4: Arm[®]

Bits 23:17 **JEDEC CODE[6:0]**: JEDEC code
0x3B: Arm[®]

Bit 16 **MEMAP**: Memory access port
1: Standard register map

Bits 15:8 Reserved, must be kept at reset value.

Bits 7:0 **IDENTITY[7:0]**: AP type identification
0x01: AHB-AP (AP0 and AP1)
0x02: APB-AP (AP2)
0x11: Reserved

60.5 Trace and debug subsystem functional description

The trace and debug subsystem features the following CoreSight components:

- System ROM tables
- Global timestamp generator (TSG)
- System cross-trigger interface (CTI)
- Cross-trigger matrix (CTM)
- Trace port interface unit (TPIU)
- Trace bus funnel (CSTF)
- Embedded trace FIFO (ETF)
- Serial wire output (SWO)
- Serial wire output trace funnel (SWTF)

These components are accessible by the debugger via the system APB-AP and its associated APB-D debug bus. They are also accessible by the Cortex-M7 processor.

The MCU debug unit (DBGMCU) is also accessed via the APB-D. This non-CoreSight component contains registers for configuring the device behavior in Debug mode.

Trace bus replicator branches the trace bus from the CPU's ITM CoreSight component to ETF and SWO, through trace bus funnels.

60.5.1 System ROM tables

There are two ROM tables on the APB-D bus. The ROM table is a CoreSight component that contains the base addresses of all the CoreSight components on the APB-D bus. These tables allow a debugger to access the topology of the CoreSight components automatically.

The first table points to the second table, and to the CoreSight components located in D3 power domain: SWO, SWTF, TSG. The DBGMCU is not referenced by the table as it is not a standard CoreSight component. The table occupies a 4-Kbyte, 32-bit wide chunk of APB-D address space, from 0xE00E0000 to 0xE00E0FFC when accessed by the debugger, and from 0x5C000000 to 0x5C000FFC when accessed from the system bus.

Table 604. System ROM table 1

Address offset in ROM table	Component name	Component base address (debugger)	Component base address (system bus)	Component address offset	Size	Entry
0x000	-	-	-	-	4 Kbytes	0x00001002
0x004	-	-	-	-	4 Kbytes	0x00002002
0x008	SWO	0xE00E3000	0x5C003000	0x03000	4 Kbytes	0x00003003
0x00C	SWO funnel	0xE00E4000	0x5C004000	0x04000	4 Kbytes	0x00004003
0x010	Timestamp generator	0xE00E5000	0x5C005000	0x05000	4 Kbytes	0x00005003